

Title: MAPK Signaling Pathway Review

Introduction

The MAPK pathway plays a critical role in cell proliferation, cell survival, and differentiation.

Pathway Flow

KRAS

↓

RAF

↓

MEK

↓

ERK

Key Observations

1. KRAS activates RAF proteins.
2. RAF activates MEK kinases.
3. MEK activates ERK.
4. ERK regulates gene expression.

Clinical Relevance

MAPK pathway dysregulation is associated with pancreatic cancer, melanoma, and lung cancer.